UNIT - IV

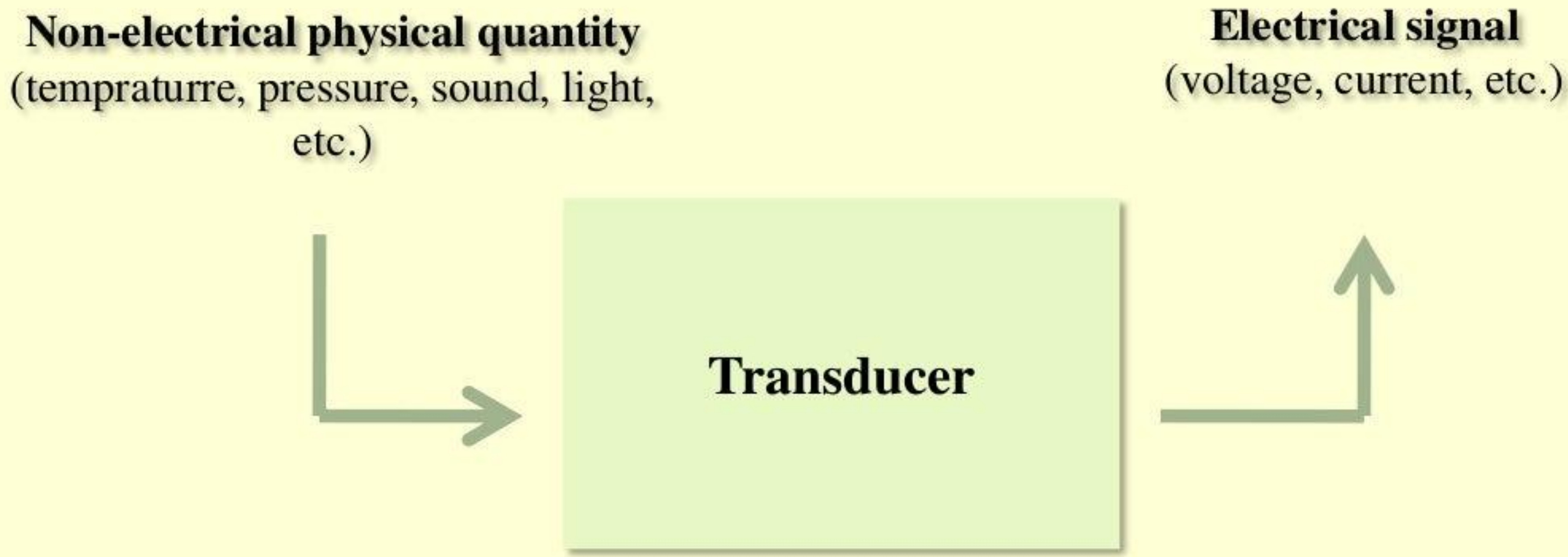
Transducers

Definition – Classification – Resistive, Inductive and Capacitive Transducer – LVDT – Strain Gauge – Thermistors – Thermocouples – Piezo electric and Photo Diode Transducers – Hall effect sensors – Numerical Problems.

Introduction

- A transducer is a device which transforms a non-electrical (physical quantity) into an electrical signal.

- Electrical and Mechanical transducer are the major types of the transducer.



Classification of Transducer based on electrical principle involved:

Variable-resistance type:

(i) **Strain and pressure gauges**

(ii) **Thermistors, resistance thermometers**

(iii) **Photoconductive cell, etc.**

Variable-inductance type:

(i) **Linear Voltage Differential Transformer (LVDT)**

(ii) Reluctance pick-up

(iii) Eddy current gauge

Variable-capacitance type:

(i) Capacitor microphone

(ii) Pressure gauge

(iii) Dielectric gauge

Voltage-generating type:

(i) **Thermocouple**

(ii) **Photovoltaic cell**

(iii) Rotational motion tachometer

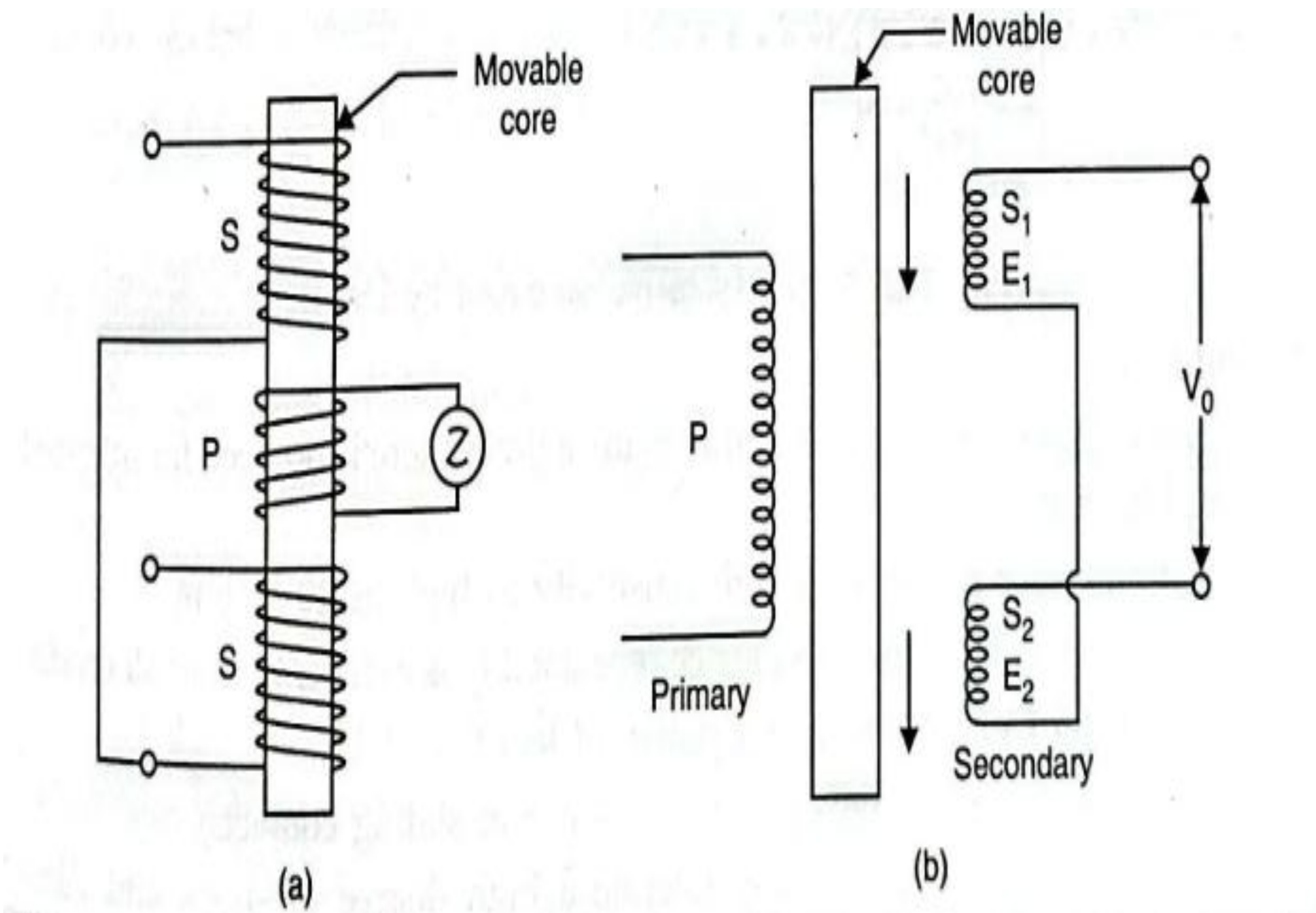
(iv) **Piezoelectric pick-up**

Voltage-divider type:

(i) Potentiometer position sensor

(ii) Pressure-actuated voltage divider

Linear Variable Differential Transducer or Transformer (LVDT):



Linear Variable Differential Transformer (LVDT):

LVDT is a passive inductive transducer and is commonly employed to measure force (or weight, pressure, and acceleration etc. which depend on force) in terms of the amount and direction of displacement of an object.

Construction

- It consists of one primary winding (P) and two secondary windings (S_1 and S_2) which are placed on either side of the primary, mounted on the same magnetic core.
- The magnetic core is free to move axially inside the coil assembly, and the motion being measured is mechanically coupled to it.
- The two secondaries S_1 and S_2 have equal number of turns but are connected in series opposition so that e.m.f.s (E_1 and E_2) induced in them are 180° out of phase with each other and hence cancel each other out.
- The primary is energized from a suitable A.C. source.

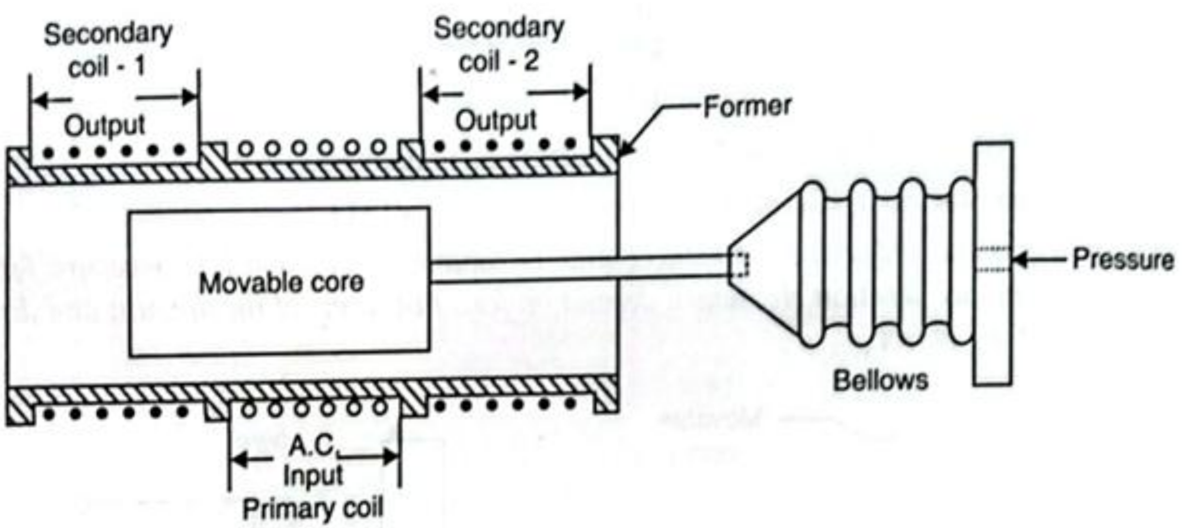
Working

When the core is in the centre (called reference position), the induced voltages E_1 and E_2 are equal and opposite. Hence, they cancel out and the output voltage V_0 is zero. When an external applied force moves the core towards coil S_2, E_2 is increased but E_1 is decreased in magnitude, though they are still antiphase with each other. The net voltage available is ($E_2 – E_1$) and is **in phase with E_2**.

Similarly, when the movable core moves towards coil S_1, $E_1 > E_2$ and $V_0 = E_1 - E_2$ and is **in phase with E_1**.

Thus, from the above discussion, the magnitude of V_0 is a **function of the distance moved by the core**, and its **polarity or phase** indicates the direction in which it has moved.

If the core is attached to a moving object, the magnitude of V_0 gives the **position of that object**.



Pressure measured by LVDT

PIEZOELECTRIC TRANSDUCERS:

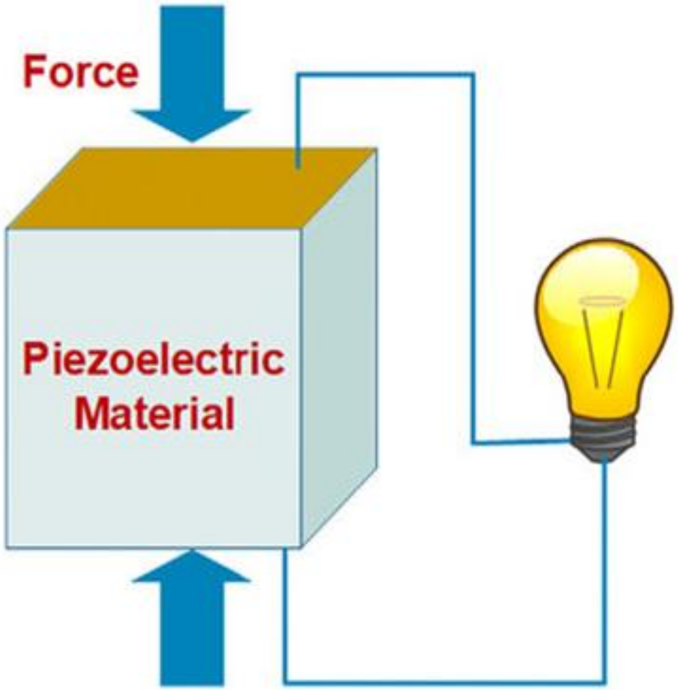
➢A piezoelectric material is one in which an electric potential appears across certain surfaces of a crystal if the dimensions of the crystal are changed by the application of a mechanical force.

This potential is produced by the displacement of external charges. The effect is reversible, i.e., conversely, if a varying potential is applied to the proper axis of the crystal, it will change the dimensions of the crystal, thereby deforming it. This effect is known as the **piezoelectric effect**.

Elements exhibiting piezoelectric qualities are sometimes known as **electro-resistive elements**. Common piezoelectric materials are:

Ammonium dihydrogen phosphate, Rochelle salts, lithium sulphate, dipotassium tartrate, potassium dihydrogen phosphate, quartz and ceramics A and B.

There are two main groups of piezoelectric crystals:
Natural crystals – such as quartz and tourmaline.
Synthetic crystals – such as Rochelle salt, lithium sulphate, dipotassium tartrate, etc.



Quartz

Tourmaline

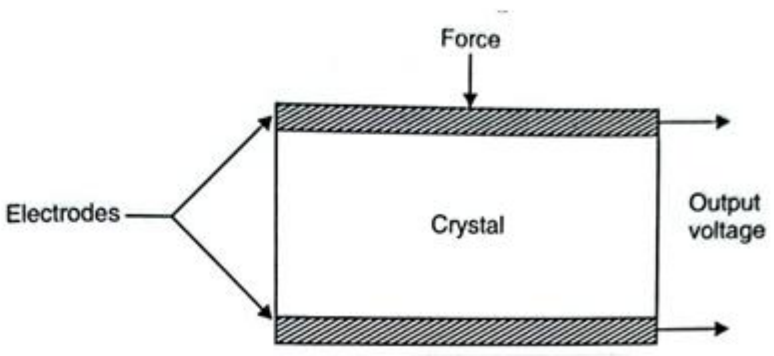
Rochelle Salts

Working of a Piezoelectric Device:

A typical mode of operation of a piezoelectric device employed for measuring varying force applied to a simple plate. The magnitude and polarity of the induced charge on the crystal surface is proportional to the magnitude and direction of the applied force. The charge at the electrode gives rise to voltage (E), given by

$$E = \frac{gtF}{A} = gtp$$

where,
g=Voltage sensitivity in Vm/N,
F=Force in N (newton),
A=Area of the crystal in m^2, and
p=F/A = Pressure in N/m^2.



Advantages:

- High frequency response.
- High output.
- Rugged construction.
- Negligible phase shift.
- Small size.

Disadvantages:

- Output affected by changes in temperature.
- Cannot measure static conditions.
- **Applications:**

These transducers find the following fields of application

- Accelerometers.

HALL EFFECT TRANSDUCERS:

Hall Effect:

Definition:

When a current carrying conductor is placed in a magnetic field, a **transverse effect** is noted. This effect is called **Hall effect** (discovered by Hall in 1879). Hall found that

"When a magnetic field is applied at right angles to the direction of electric current an electric field is set up which is perpendicular to both the direction of electric current and the applied magnetic field."

- ➢The **Hall effect element** is a type of **transducer** used for **measuring the magnetic field** by **converting it into an emf (voltage)**.
- ➢Since **direct measurement** of magnetic field strength is not possible, the **Hall Effect Transducer** is used to obtain an electrical signal proportional to the field.
- ➢This **transducer converts** the **magnetic field** into an **electrical quantity**, which can be easily measured using **analog** or **digital meters**.

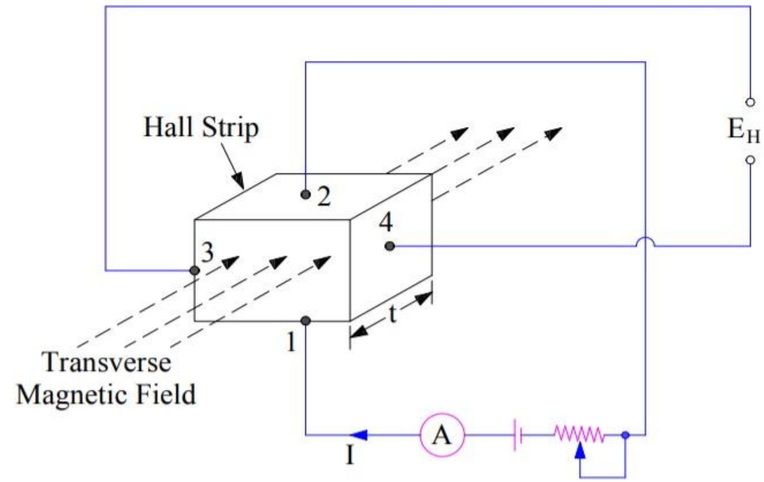
$$V_H = \frac{R_H B I}{b}$$

R_H = Hall coefficient,

B = Magnetic field strength,

I = Current carried by the specimen, and

b = Width of the specimen along the magnetic field.



Working of Hall Effect Transducer:

- The **current is supplied** through **leads 1 and 2**, and the **output voltage** is obtained from **leads 3 and 4** of the Hall element.
- When **no magnetic field** is applied, the **leads 3 and 4** remain at the **same potential** — hence, no output voltage is developed.
- When a **magnetic field** is applied **perpendicular** to the current flow in the strip, an **output voltage (Hall voltage)** is developed across **leads 3 and 4**.
- The **developed voltage** (Hall voltage) is **directly proportional** to the **magnetic field strength** and depends on the **properties of the material** used.
- The **principle** of the Hall effect transducer is that **when a current-carrying conductor or strip** is placed in a **transverse magnetic field**, an **emf (Hall voltage)** is **developed across the edges** of the conductor. The **magnitude of this developed voltage** is **directly proportional** to the **magnetic flux density (B)**. This phenomenon is known as the **Hall effect**.
- The **Hall effect element** is mainly used for **measuring magnetic fields** and for **sensing current** in electrical and electronic systems.

The following are the applications of Hall effect transducers:

1. Displacement measurement:

- The **Hall effect element** is used to **measure the displacement** of a **structural or mechanical component**.
- The **Hall effect transducer** is placed **between the poles of a permanent magnet**.
- When the **ferromagnetic plate** moves (due to displacement), it **changes the magnetic field strength** across the Hall element.
- This **variation in magnetic field** causes a **change in Hall voltage**, which is **proportional to the displacement** of the ferromagnetic plate.

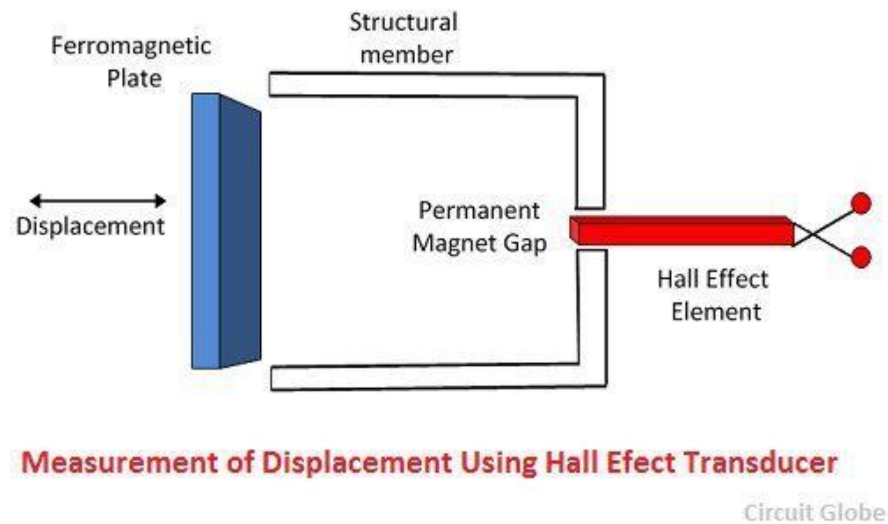


Fig. Hall effect Displacement transducers

2. Measurement of Current

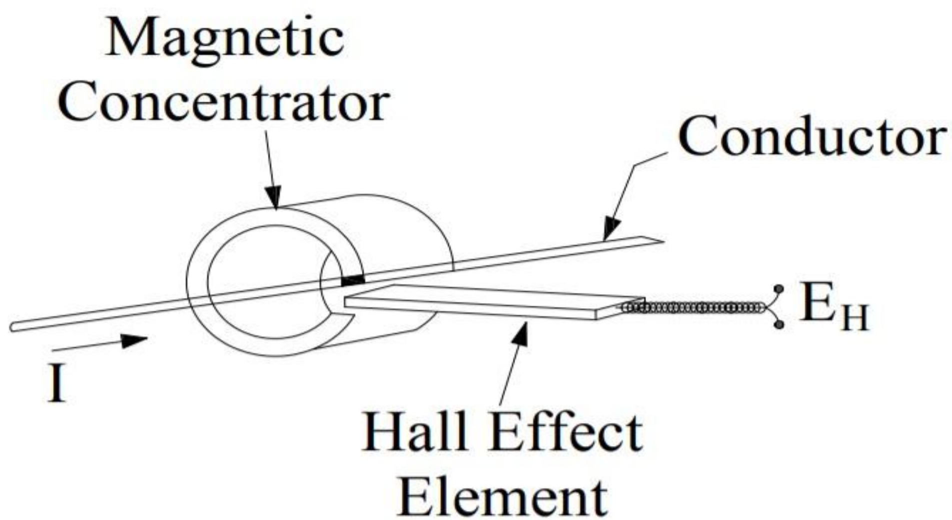
➢The **Hall effect transducer** is used for **measuring current without any physical connection** between the conductor and the measuring circuit.

➢**AC or DC current** flowing through the conductor produces a **magnetic field** around it.

➢The **strength of the magnetic field** is **directly proportional** to the **current** flowing through the conductor.

➢When a **Hall effect element** is placed **perpendicular** to this magnetic field (inside a **ferromagnetic concentrator**), it develops a **Hall voltage (E_h)**.

➢The **Hall voltage** produced is **proportional to the current** in the conductor and follows the **Biot–Savart Law**.



3. Measurement of Power

- The **Hall effect transducer** is used for **measuring the power** of a conductor.
- The **current** flowing through the conductor produces a **magnetic field** around it.
- The **intensity of the magnetic field** is **directly proportional** to the current.
- This magnetic field induces a **Hall voltage (E_h)** across the **Hall element**.
- The **output voltage of the multiplier** circuit is **proportional to the power** being measured by the transducer.

Strain Gauge Transducer:

What is a Strain Gauge?

➢A strain gauge is a resistor used to measure the strain (deformation) on an object.

➢When an external force is applied to an object, a change in shape or deformation occurs.

➢This deformation may be compressive or tensile, and the resulting strain is measured by the strain gauge.

➢When the object deforms within its elastic limit, it either becomes narrower and longer or shorter and broader.

➢As a result, there is a change in resistance of the strain gauge.

➢By measuring this change in resistance, the induced stress or strain can be accurately determined.

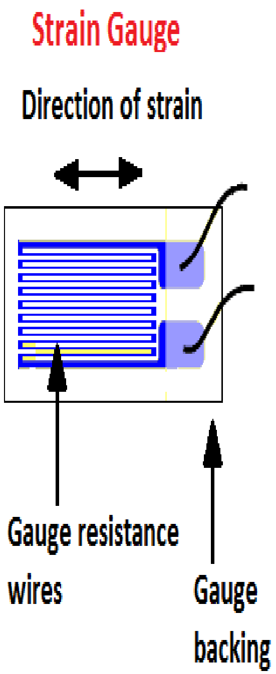


Figure #1

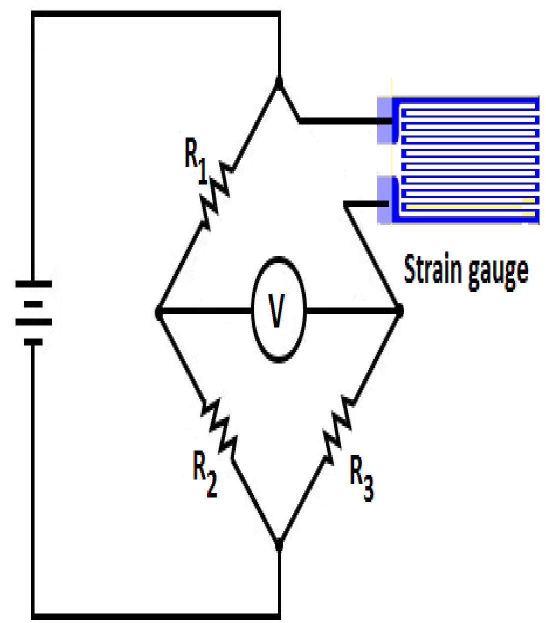


Figure #2

How does it work?

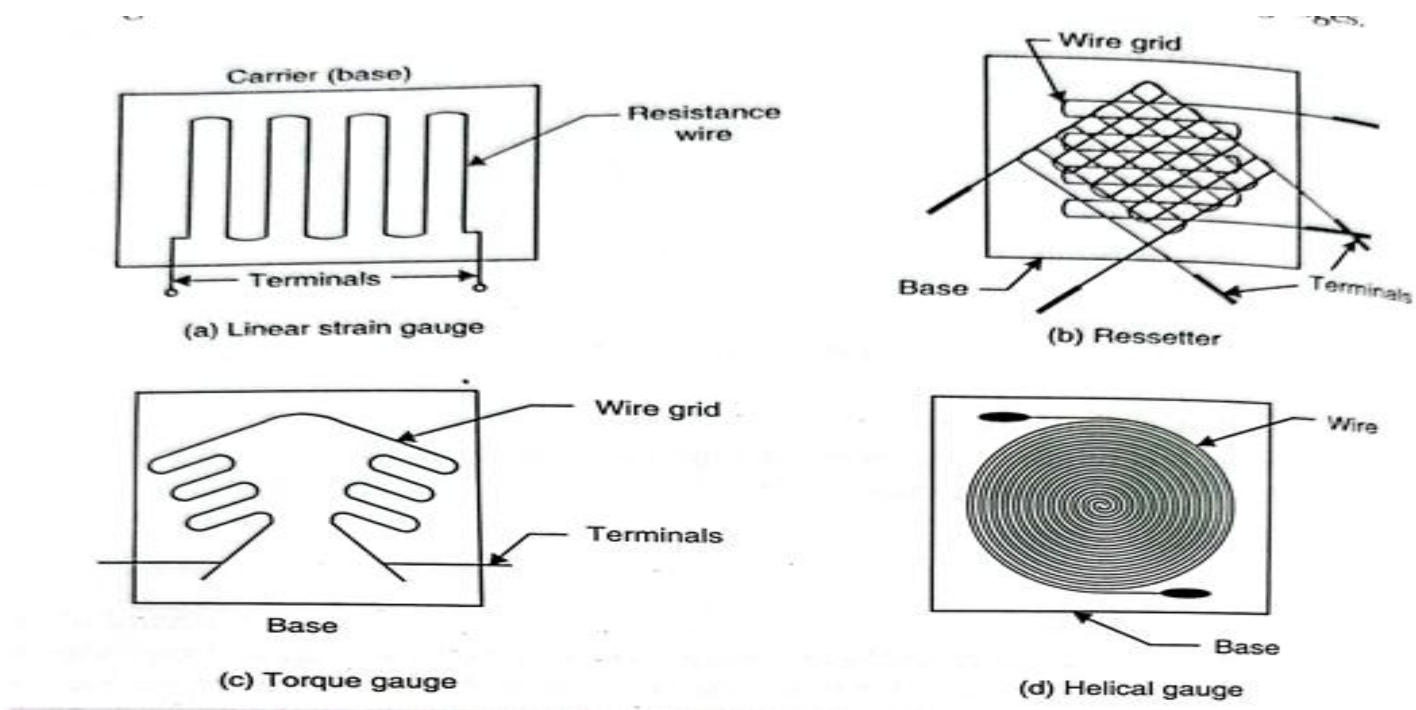
➢In this circuit, **R_1 and R_3** are the ratio arms equal to each other, and **R_2 is the rheostat arm** having a value equal to the strain gauge resistance.

➢When the gauge is unstrained, the bridge is **balanced**, and the **voltmeter shows zero value**.

➢As there is a change in the resistance of the strain gauge, the bridge gets **unbalanced** and produces an **indication at the voltmeter**.

➢The output voltage from the bridge can be amplified further by a **differential amplifier**.

Classification of Strain Gauge Transducer:



Advantages of Strain Gauge Transducer:

•**High Accuracy:**
Provides precise measurement of strain, stress, and force.
•**Compact and Lightweight:**
Easy to attach to small or delicate structures.
•**Wide Measurement Range:**
Can measure both very small and large strains accurately.
•**Fast Response:**
Suitable for dynamic and transient measurements.
•**High Sensitivity:**
Small changes in strain produce noticeable resistance changes.
•**Simple and Robust Construction:**
Can work in harsh industrial environments.
•**Can Be Used in Bridges:**
Works effectively in Wheatstone bridge circuits for accurate signal conditioning.

Disadvantages of Strain Gauge Transducer:

•**Temperature Sensitive:**
Changes in temperature affect resistance and cause measurement errors.
•**Requires Signal Conditioning:**
Output is very small and needs amplification.
•**Difficult Installation:**
Needs careful surface preparation and bonding.
•**Limited Strain Range:**
Cannot measure very large deformations (may get damaged).
•**Hysteresis and Creep:**
Errors may occur due to long-term loading or repeated use.
•**Requires Calibration:**
Needs frequent recalibration for accuracy.